

Final Report E50 Voice Command Machine

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Program name	E50 Voice Command Machine, final frozen Raspberry Pi deployment
Final experiment	E50_BG_REVISIED_VOCAB_ORIGINAL_PRESERVE_E41_INIT
Target hardware	Raspberry Pi 5
Repository package	GITHUB_PACKAGE_FINAL_20261002
Command model SHA256	BC8AC64CED4305DDF43BF4DE377F3CC636A764EFF339CD9F92AF06340C50B8FF
Wake model	E37_TARGETED_COLOR_VOLUME_FIX
Threshold policy	E40_NON_LED_COMMAND_GUARDRAIL_THRESHOLDS

This report is the technical summary of the final E50 evidence. It documents the E50 implementation, model, dataset, thresholds, router, response assets, runtime, GUI boundary, benchmark evidence, repository package, verified Demo Day metrics, benchmark audit, post-freeze GUI clarification, and independent E53 documentation.

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1 Executive Summary

E50 is the final Machine Exercise 2 Voice Command Machine delivery: an offline Raspberry Pi 5 voice-command system with a frozen 19-label command vocabulary, a separate wake gate, a command CNN, a confidence/rejection policy, deterministic local routing, local response audio, touchscreen GUI operation, and persistent return-to-listening behavior.

The final frozen recognition/action stack is:

```
Pi microphone -> E37 wake -> command capture -> log-Mel preprocessing -> E50
command CNN -> E40 guardrail -> deterministic router -> local action -> local WAV
response -> return to listening
```

The final physical Raspberry Pi benchmark used 20 trials: 19 valid command-label trials plus one UNKNOWN/no-action trial. Wake success was $19/20 = 95.0\%$ overall and $18/19 = 94.74\%$ for valid command trials. Raw command classification was $13/19 = 68.42\%$. End-to-end action success was $10/19 = 52.63\%$. All accepted/executed valid commands produced the correct action, giving accepted-action precision of $10/10 = 100\%$. All non-executed trials were safely rejected without a routed action, giving safe rejection of $10/10 = 100\%$. The UNKNOWN/no-action trial was safe at $1/1 = 100\%$. The runtime returned to listening after all 20 trials.

These metrics deliberately remain separate. The report does not collapse them into a single “overall accuracy” number. Raw classification, confidence acceptance, action execution, safe rejection, and end-to-end success measure different layers of the system.

The final system demonstrates strong action safety, conservative rejection behavior, fast local inference, offline operation after setup, deterministic routing, and reproducible frozen identity. Its main limitation is command-level recognition coverage across the full vocabulary: raw classification was moderate at $13/19$, and end-to-end action success was $10/19$ in the final physical benchmark.

This report also preserves project boundaries. E53/VCM2 is an independent experiment. It is not an E50 model, training stage, benchmark, deployment component, improvement, or replacement. No E53 implementation file, model, dataset, configuration, or evidence artifact is part of the frozen E50 system.

2 Project Identity and Final Frozen State

Field	Final E50 value
Project	ME2 Voice Command Machine, final E50 Raspberry Pi delivery
Final experiment ID	E50_BG_REVISED_VOCAB_ORIGINAL_PRESERVE_E41_INIT
Target hardware	Raspberry Pi 5
System type	Offline Raspberry Pi 5 voice-command system for fixed command labels and local actions
Wake model	E37_TARGETED_COLOR_VOLUME_FIX
Wake threshold	0.90
Command model	E50_BG_REVISED_VOCAB_ORIGINAL_PRESERVE_E41_INIT
Command model SHA256	BC8AC64CED4305DDF43BF4DE377F3CC636A764EFF339CD9F92AF06340C50B8FF

Threshold policy	E40_NON_LED_COMMAND_GUARDRAIL_THRESHOLDS
Command default threshold	0.90, with documented label-specific threshold behavior
Runner	scripts/run_vcm_touchscreen_gui.sh
GPIO	Disabled in final delivery
Response	Local WAV/action response enabled
Final vocabulary size	19 labels

The final E50 vocabulary is:

1. PLAY_MUSIC
2. WEATHER
3. TIME
4. LIGHT_ON
5. LIGHT_OFF
6. LIGHT_DIM
7. BRIGHTNESS
8. TIMER
9. ALARM
10. TEMPERATURE
11. NEXT
12. PAUSE
13. STOP
14. VOLUME_UP
15. VOLUME_DOWN
16. CREATE_REMINDER
17. LIST_REMINDERS
18. CALL
19. MESSAGE

Important vocabulary boundary: COLOR is not a final E50 class. The final E50 system uses LIGHT_DIM. Any evidence involving COLOR belongs to an earlier experiment, collective benchmark schema, or historical command-design context unless explicitly identified otherwise.

3 Deployment Scope and Objectives

3.1 Scope

The deployed E50 system covers:

- Raspberry Pi 5 runtime and GUI launcher.
- E37 wake detection.
- E50 command-recognition CNN.
- E40 confidence/rejection policy.
- 19-label command vocabulary.
- Deterministic router/action layer.
- Local response WAV assets and audio playback.
- Offline operation after setup.
- Validation, benchmark, integrity, reproducibility, and technical documentation.

The system is not a cloud assistant, not an LLM, not an online ASR service, and not an arbitrary transcription system. It is a wake-gated fixed-vocabulary command classifier with deterministic local action routing.

3.2 Objectives

1. Build a local Raspberry Pi voice-command machine that can wake, classify commands, route actions, play responses, and return to listening.
2. Preserve the complete final 19-label vocabulary without a runtime whitelist.
3. Demonstrate offline operation without cloud or network inference during verified operation.
4. Validate safety behavior for unsupported, silent, wrong, uncertain, or below-threshold input.
5. Report benchmark metrics by layer: wake, raw classification, acceptance, action correctness, safe rejection, end-to-end success, and runtime recovery.
6. Package the project so a technical reader can inspect requirements, architecture, implementation, engineering history, validation, limitations, reproduction steps, and integrity records.

4 Architecture

Stage	Role in E50
Pi microphone	Captures physical audio input.
E37 wake model	Detects the wake phrase and gates command capture.
Command capture	Records the command window after wake acceptance.
Log-Mel preprocessing	Converts captured command audio into fixed features.
E50 command CNN	Predicts one of the 19 final command labels.
E40 policy	Accepts or rejects predictions using frozen confidence thresholds.
Router/action layer	Maps accepted labels to deterministic local actions.
Local response WAV	Plays the corresponding local response audio.
Return to listening	Restores persistent runtime state after each trial.

The GUI is an interface and launcher. Recognition and routing are runtime responsibilities downstream of the frozen E37/E50/E40 stack.

4.1 Recovered E37 Wake-Recording Provenance

Recovered E37 provenance evidence documents the original project-specific Raspberry Pi Hey Pi wake-recording resource. The recovered manifest is `/home/loreenanne/vcm_pi_package/pi_validation/wake_validation_hey_pi/manifest.csv`. It contains 99 wake-stage recordings: 50 WAKE / hey pi rows, 15 UNKNOWN rows, 14 COLOR targeted rows, 14 VOLUME_UP targeted rows, 3 LIGHT_ON command-as-wake-probe rows, and 3 PLAY_MUSIC command-as-wake-probe rows. The manifest records 64 adaptation-designated rows and 35 holdout-designated rows. All rows are documented as 4-second, 16 kHz recordings using input device `plughw:2,0`.

This evidence establishes the existence and composition of the E37-related Hey Pi recording set. It does not independently establish the exact final E37 training-row membership, speaker/recording count, training epochs, optimizer, validation metrics, augmentation count, or row-level mapping from the 99 recordings to the final E37 checkpoint. The E37 wake-recording lineage is also separate from the collective Gold Dataset lineage used for the E50 command model.

5 Dataset and Training Provenance

5.1 Collective Gold Dataset Lineage

E50 was developed using selected material from the class collective Gold Dataset:

GoogleDrive <https://drive.google.com/drive/folders/1k2RklcsU8ull9-4fagl-8zTfecPaQWBZ>

HuggingFace <https://huggingface.co/datasets/airimonda/ai231-me2-voice-commands>

Documentation <https://claude.ai/artifact/PPNHMWd5rx9qcXTdcukV7s> rather than using the entire collective dataset unchanged. The project-specific dataset was further prepared and balanced for the 19-command E50 vocabulary, with training-only augmentation and a separate Pi deployment-adaptation component. The resulting dataset therefore retains lineage to the collective dataset while incorporating project-specific engineering data.

The final E50 training manifest contains 16,100 rows: 13,070 active-project rows, 2,800 selected VCM_BALANCED rows, and 230 separately identified E41 Pi adaptation rows. The first two components are collective-derived/project-local command-data branches; the E41 rows are project-specific deployment adaptation data and should not be described as Gold Dataset rows unless additional evidence establishes that link.

VCM_MASTER is a project-local Dataset2 / collective-family representation derived from selected collective dataset material; it is not claimed to be the entire Gold Dataset. VCM_BALANCED is a training-derived subset of VCM_MASTER and therefore inherits its collective-family provenance. E53/VCM2 remains an independent experiment and is not merged into E50.

5.2 Dataset A: VCM_MASTER

Item	Value
Dataset name	VCM_MASTER
Total samples	36,622
Classes	16
Training samples	27,130
Validation samples	4,734
Test samples	4,758
Speaker/group identities	395
Speaker leakage	Zero train/validation/test overlap documented

5.3 Dataset B: VCM_BALANCED

Item	Value
Dataset name	VCM_BALANCED
Training samples	15,268
Original samples	13,801
Augmented samples	1,467
Relationship to Dataset A	Derived from Dataset A training material; did not introduce validation/test leakage

5.4 Final E50/BG Training Manifest: 16,100 rows

Source	Rows
active_project_dataset	13,070
dataset2_vcm_balanced	2,800
e41_reconstructed_adaptation	230

The final data-kind composition was:

Kind	Rows
Real	15,260
Augmented	480
Real multi-sensor	200
Synthetic	160
Total	16,100

The final distribution was controlled rather than perfectly balanced. The engineering rule retained E41 lineage where applicable, capped active-project original data at 700 rows per label, and added up to 200 Dataset2 rows where usable support existed. This yielded 14 labels with 900 rows and five labels with 700 rows. The five 700-row labels were BRIGHTNESS, CREATE_REMINDER, LIST_REMINDERS, CALL, and MESSAGE.

This is best described as controlled class balancing with bounded auxiliary-data addition. Synthetic and augmented data are training-domain support, not independent real-speaker deployment evidence.

5.5 Training History

The final selected checkpoint was epoch 7, selected by best internal validation loss.

Epoch	Train loss	Train acc.	Val. loss	Val. acc.	Status
7	0.3458	0.8801	1.3988	0.6197	Best / selected
11	0.1946	0.9322	2.1883	0.6018	Later / not selected

The later training trajectory shows a classical overfitting pattern: training loss decreased and training accuracy increased, while validation loss worsened and validation accuracy did not improve meaningfully. However, this does not mean the deployed E50 checkpoint was a late overfit model. Epoch 7 was the selected checkpoint; later overfit epochs were not deployed. Deployment failures should therefore be attributed by evidence-supported layer rather than blamed entirely on overfitting.

6 Execution Timeline and Engineering Process

The project followed an evidence-gated engineering process rather than a single uncontrolled training run. Earlier experiments diagnosed model, threshold, router, action, audio-output, and deployment issues separately. Candidate models and recovery experiments were rejected when they introduced regressions or lacked adequate independent validation.

The final engineering process preserved the following principles:

- keep model development separate from final validation;
- keep wake detection, command classification, confidence policy, routing, action execution, response playback, and GUI behavior separable in evidence;

- avoid post-hoc threshold tuning against final benchmark evidence;
- preserve all 19 final labels as callable/testable;
- report demo readiness separately from runtime callability;
- document limitations instead of patching the frozen system after benchmark closure.

The project record documents major September 30 closure activities: offline operation evidence, safety sweep evidence, Pi performance measurement, all-19 final benchmark evidence, and final artifact integrity work. This report presents that project evidence together with the clarified timing, benchmark-audit, GUI-boundary, and E53-separation language used in the final submission package.

7 Final Benchmark Study

7.1 Benchmark Population

The final physical Raspberry Pi benchmark population was:

- 19 valid command-label trials, one for each final E50 label;
- 1 UNKNOWN/no-action trial;
- 20 total physical Pi 5 trials.

The benchmark recorded wake detection, raw predicted label, command confidence, threshold, acceptance, expected action, actual action, action correctness, response playback, return-to-listening, end-to-end success, and failure layer.

7.2 Final Pi Results

Metric	Result	Meaning
Wake success	19/20 = 95.0%	Wake accepted across all benchmark trials.
Wake success on valid commands	18/19 = 94.74%	Wake accepted for valid command-label trials.
Raw command classification	13/19 = 68.42%	Correct raw predicted command label among valid commands.
Accepted valid commands	10/19 = 52.63%	Valid commands accepted by E40 policy.
Accepted-correct	10	Accepted command produced correct corresponding action.
Accepted-wrong	0	No accepted command produced a wrong action.
Accepted-action precision	10/10 = 100%	Correct action among accepted/executed valid commands.

Safe rejection among non-executed	10/10 = 100%	Non-executed trials produced no routed action.
End-to-end action success	10/19 = 52.63%	Valid command trial completing the defined successful E50 action path, including correct command outcome, accepted execution, response handling, and return-to-listening.
UNKNOWN/no-action safety	1/1 = 100%	Unsupported phrase produced no action.
Return-to-listening	20/20 = 100%	Runtime returned to listening after every trial.

The distinction matters. Raw command classification measures the CNN label. Acceptance measures the E40 policy. Accepted-action precision measures whether accepted commands routed to the correct action. Safe rejection measures conservative no-action behavior. End-to-end action success requires the complete successful path.

7.3 Observed Failure Layers

The final Item 16 per-trial data show that unsuccessful valid-command trials were not downstream action-routing bugs. They included wrong raw predictions safely rejected, correct raw predictions rejected below threshold, and one wake rejection. Accepted wrong actions were zero. This supports the conclusion that the final benchmark failures were recognition, wake, or confidence/rejection failures rather than erroneous executed actions.

8 Performance and Efficiency

8.1 Model Size and Computational Cost

Metric	Value	Definition
Parameters	66,483	Final E50 command CNN parameters.
Weights	251,734 bytes	Final E50 weights artifact size.
Weights + normalization	252,464 bytes	Weights plus normalization artifact.
MACs	85,774,144	Conv2D/Dense MACs per 4-second input.
Pi model load	1.8566 s	Model load time in Item 15 benchmark.

8.2 CNN Inference Latency and CNN-Only RTF

Measurement	Value	Boundary
Mean CNN inference	31.82 ms	CNN inference only.
P50 CNN inference	31.39 ms	CNN inference only.
P95 CNN inference	34.40 ms	CNN inference only.
Maximum CNN inference	38.31 ms	CNN inference only.
CNN-only RTF mean	0.0080	CNN inference time / 4-second command input.
CNN-only RTF P95	0.0086	CNN inference time / 4-second command input.

CNN-only RTF is not full end-to-end RTF, wake-to-action RTF, wake-to-response RTF, or acoustic response latency.

8.3 Post-Closure Response Playback-Start Timing

A post-closure passive timing audit measured local response playback-start latency without modifying the frozen runtime. The measurement population was five accepted/executed trials. The boundary was:

command WAV capture completion -> first observed local aplay response-WAV process

Measurement	Value	Qualification
N	5 accepted/executed trials	Qualified post-closure passive audit.
Mean playback-start latency	66.172 ms	Not acoustic speaker-onset.
P50 playback-start latency	64.837 ms	Not full microphone-to-speaker latency.
P95 playback-start latency	73.999 ms	Not full wake-to-response latency.
P99 playback-start latency	73.999 ms	Same measurement boundary.
Maximum playback-start latency	73.999 ms	Same measurement boundary.
Command-pipeline RTF mean	0.016543	Capture completion to response playback-start / 4 s.
Command-pipeline RTF P95	0.018500	Capture completion to response playback-start / 4 s.

This measurement strengthens the timing evidence, but it does not establish acoustic speaker-onset latency, full microphone-to-speaker latency, full wake-to-action latency, or full wake-to-response latency.

9 Safety, FAR, and Failure Analysis

9.1 Bounded FAR

The established false-accept result is:

Bounded command-window FAR: $0/4 = 0.0\%$

The definition is: false accepts among valid out-of-scope command-window safety prompts, where a false accept means the frozen E50 system accepted and executed a command.

This is a bounded command-window FAR. It is not a broad environmental false-accept rate, universal false-wake rate, or proof of robustness to arbitrary background audio. Broad environmental FAR was not established by the available evidence.

9.2 Safety Findings

The final benchmark supports these safety conclusions within the tested population:

- accepted-wrong actions: 0;
- accepted-action precision: $10/10 = 100\%$;
- safe rejection among non-executed trials: $10/10 = 100\%$;
- UNKNOWN/no-action safety: $1/1 = 100\%$;
- return-to-listening: $20/20 = 100\%$.

These results support a conservative safety interpretation: uncertain or wrong recognition was generally converted into rejection instead of wrong device action. They do not support the claim that command recognition was near-perfect.

9.3 Failure Attribution

The final report does not use overfitting as a catch-all explanation. Evidence-supported contributing factors include:

- classifier confusion between acoustically or representationally similar classes;
- confidence rejection of correct but low-confidence predictions;
- wake rejection in one final valid-command trial;
- dataset coverage limitations for selected labels;
- speaker, acoustic, phrasing, and capture-window effects;
- training dynamics after epoch 7;
- domain shift between internal validation, saved-WAV benchmarks, and live Pi operation.

No accepted final benchmark command routed to an incorrect action. Therefore the final benchmark does not support a claim of router/action implementation bugs in accepted commands.

10 Comparable Baseline

E51 is the comparable-size offline baseline. It has the same parameter count as E50 but is not a second physical-Pi benchmark.

Metric	E50		E51 Interpretation
Parameters	66,483		66,483 Comparable size.
Current95-compatible	83/90 = 92.22%	77/90 = 85.56%	E50 higher on this offline set.
Phase-AV-compatible	139/194 = 71.65%	131/194 = 67.53%	E50 higher on this offline set.
Combined revised	2724/4230 = 64.40%	2727/4230 = 64.47%	Essentially tied, E51 slightly higher by count.
Phase-AV accepted precision	91.92%	89.61%	E50 higher on this metric.

The comparison should be interpreted as a qualified offline same-architecture baseline, not as proof that E50 dominates all alternatives and not as a second final Pi deployment result.

11 GUI Enhancement After Core Freeze

11.1 Frozen Core

The following remained frozen for final benchmark identity:

- E50 command model;
- E37 wake model;
- E40 thresholds;
- command vocabulary;
- router/action logic;
- response assets;
- core recognition behavior;
- benchmarked system identity.

11.2 Post-Freezing GUI Work

The GUI work was a post-freeze presentation and runtime-control enhancement around the frozen E50 VCM core. It did not retrain E37 or E50, tune E40 thresholds, change the final vocabulary, modify router/action semantics, replace response assets, or alter the benchmarked recognition/action path. Instead, the touchscreen GUI launches the existing wake-gated runtime as a subprocess, using the frozen E37 wake model, the E50 command model, the E40-compatible threshold policy, the existing response-audio map, and the continuous return-to-listening runtime.

The GUI provides an operator-facing control and display layer. It presents large state prompts such as WAKE ME UP, SAY WHAT YOU NEED, I AM NOT AWAKE, I DIDN'T UNDERSTAND, and I CAN'T DO THAT; start/stop controls for the VCM runtime; and live fields for the last predicted command, routed action, response playback status, and offline system status. It reads runtime output and the latest result JSON files to update the display after wake acceptance, wake rejection, command execution, below-threshold rejection, unsupported command rejection, and return-to-listening behavior.

The telemetry enhancement adds run-level operator diagnostics without changing model decisions. During a GUI-controlled run, it counts wake detections, wake rejections, recognized/executed commands, below-threshold command rejections, and unsupported/not-allowed command outcomes. These counts are displayed as live run statistics with count/total percentages, and transaction records are appended to `runtime_diagnostics/vcm_run_statistics.jsonl`. This telemetry is useful for usability review, operator feedback, and runtime diagnostics, but it is not part of the E50 benchmark evidence and is not evidence of improved model accuracy. The benchmarked system identity remains the frozen E37/E50/E40 recognition-action core; the GUI improves presentation, control, and observability around that core.

12 Independent E53 / VCM2 Experiment

E53/VCM2 is a separate experimental track documented in the technical package. It is not part of the final E50 implementation or benchmark.

12.1 What E53 Tested

The E53 investigation asked whether a separate offline non-ASR voice-command classifier could be developed from an independently organized dataset and controlled experiment sequence. The documented E53 dataset, `E53_DATASET_V1`, contained 24,706 rows across 20 classes: 18 operational commands plus UNKNOWN and SILENCE. The split recorded 19,553 training rows, 2,544 validation rows, and 2,609 test rows. E53 investigated small CNN classifiers and representation variants including log-Mel, MFCC, and PCEN.

12.2 Documented E53 Findings

The documented LOG011 control produced test accuracy 0.463779, macro F1 0.410447, and weighted F1 0.445472. UNKNOWN F1 was 0.922756 and SILENCE F1 was 0.585242.

LOG025 compared MFCC and PCEN representations:

- MFCC: test accuracy 0.478727, macro F1 0.447684, weighted F1 0.450891, disposition mixed / inconclusive.
- PCEN: test accuracy 0.463779, macro F1 0.387960, weighted F1 0.449258, disposition mixed / inconclusive.

E53 documented unresolved issues, including command-pair confusions such as `LIST_REMINDERS/CREATE_REMINDER` and `VOLUME_UP/VOLUME_DOWN`, and tradeoffs between operational-command behavior and UNKNOWN/SILENCE behavior.

12.3 Why E53 Did Not Become E50

E53 did not establish final Raspberry Pi deployment, GUI operation, router/action integration, response playback, network-off standalone operation, Section 68 completion, or replacement of E50. It was not used in the final E50 implementation, model stack, benchmark, validation evidence, runtime, threshold policy, router, response assets, or GitHub delivery implementation. E53 did not beat or replace E50 and was not production-ready.

E53 is therefore reported as independent research evidence only. It provides useful lessons about dataset integrity, provenance, speaker/source grouping, safety/background classes, representation choice, and class-pair separability. It does not change E50 metrics.

13 Benchmark Study and Class Benchmark Adaptation

The class/collective VCM benchmark methodology (<https://github.com/airimonda/vcm-benchmark>) was reviewed for compatibility with E50. The broader benchmark uses a 19-intent/93-command-variation structure, includes slot metrics, and includes COLOR. E50 is instead a fixed-label 19-command classifier with LIGHT_DIM, native E50 JSON evidence, E40 confidence policy, and deterministic routing.

E50 adopted compatible benchmark concepts:

- wake metric;
- raw command classification;
- acceptance and rejection behavior;
- accepted-action precision;
- safe rejection;
- UNKNOWN/no-action safety;
- return-to-listening;
- bounded FAR;
- latency and RTF, with explicit boundaries;
- comparable-size baseline.

E50 did not import incompatible metrics merely for completeness. Slot-value exact match, slot MAE/MRE, phonetic distance, character distance, arbitrary unknown-intent-name metrics, and 93-command statistics are not directly applicable to final E50.

14 Requirements Fulfillment

Requirement	Evidence	Status
Model	E50 identity, SHA, architecture, parameter count, weights, MACs.	PASS
Dataset	Dataset A/B provenance, final E50 manifest, speaker separation.	PASS
Training	Training history, selected checkpoint, internal validation evidence.	PASS
Pi validation	Final Item 16 physical benchmark, 20 trials.	PASS
Keyword/intent accuracy	Wake success and raw classification reported separately.	PASS
False-accept rate	Bounded command-window FAR 0/4; broad environmental FAR not established.	QUALIFIED
Latency	CNN P95 34.40 ms and playback-start P95 73.999 ms with boundaries.	QUALIFIED
RTF	CNN-only RTF and qualified command-pipeline RTF; not full acoustic E2E RTF.	QUALIFIED
Runtime	Offline Pi local runtime, deterministic routing, response audio, return-to-listening.	PASS
Public/reproducible package	GitHub package and deployment package documented.	QUALIFIED
Dataset citation/license	Upstream source terms documented; unrestricted redistribution not established.	QUALIFIED
Training logs/checkpoint	Present in project evidence.	PASS
Pi latency reproducibility	Item 15 benchmark and passive timing audit documented.	QUALIFIED
Held-out/unseen speakers	Dataset speaker split established; formal Pi unseen-speaker benchmark not established.	QUALIFIED
Comparable baseline	E51 same-parameter offline baseline.	QUALIFIED

15 Evidence Boundaries and Limitations

The following were not established by the available evidence:

- broad environmental FAR;
- formal statistically characterized unseen-human-speaker physical-Pi benchmark;
- full wake-to-action latency;
- full wake-to-response latency;
- acoustic speaker-onset latency;
- full acoustic end-to-end RTF;
- ECE/calibration;
- broad noise/reverberation robustness;
- systematic phrasing-variation robustness;
- statistically meaningful per-intent metrics from the one-trial-per-label final benchmark;
- unrestricted dataset redistribution rights;
- final E50 metric using `COLOR` as a final class.

Dataset-level speaker separation was established: 395 identities and zero train/validation/test speaker overlap. That is strong evidence for the dataset split, but it is not the same as a formal statistically characterized unseen-human-speaker physical-Pi benchmark.

16 Integrity and Project Separation

The final integrity position is:

- E50 model unchanged after freeze;
- E37 unchanged;
- E40 unchanged;
- no post-freeze retraining;
- no post-freeze threshold tuning;
- no command whitelist;
- no benchmark manipulation;
- GUI enhancement did not modify the frozen recognition/action core;
- E53 remained independent;
- benchmark results were not replaced with more favorable results;
- limitations are reported rather than hidden.

This report is documentation only. It does not alter the frozen system or evidence.

17 Final Findings

1. E50 is a functioning offline Raspberry Pi 5 VCM final-delivery system.
2. The frozen stack is traceable: E37 wake, E50 command CNN, E40 policy, deterministic routing, local response audio, and return-to-listening.
3. The final physical benchmark tested all 19 final command labels plus one UNKNOWN/no-action trial.
4. Wake behavior was strong in the tested population: 19/20 overall and 18/19 valid-command trials.
5. Raw command recognition remains moderate: 13/19 valid commands.
6. End-to-end action success remains moderate: 10/19 valid commands.
7. Safety behavior was strong in the tested population: accepted-wrong actions were zero, accepted-action precision was 10/10, safe rejection was 10/10, UNKNOWN safety was 1/1, and return-to-listening was 20/20.
8. CNN inference was fast on Pi 5, with P95 34.40 ms and CNN-only P95 RTF 0.0086.
9. Post-closure passive timing established qualified playback-start latency, but not full acoustic response latency.
10. E51 supplies a comparable-size offline baseline, not a second Pi benchmark.
11. E53 is independent research evidence and did not modify or replace E50.

18 Final Summary

E50 demonstrates an offline, local, wake-gated command system with verified Raspberry Pi deployment, a frozen 19-label vocabulary, deterministic action routing, local response playback, and traceable final benchmark evidence. Its strongest demonstrated properties are action safety, conservative rejection, fast local inference, reproducible frozen identity, and runtime recovery.

Its principal demonstrated limitation is command-recognition coverage. The model did not correctly classify every command in the final physical benchmark, and end-to-end action success covered 10 of 19 valid commands. The system is therefore best characterized as a conservative voice-command prototype: uncertain or wrong recognition was generally converted into no-action behavior, rather than into wrong device actions.

19 Final Conclusion

E50 constitutes the completed final-delivery implementation for Machine Exercise 2, with the documented validation results and limitations described in this report. It is a functioning offline Raspberry Pi VCM with frozen recognition artifacts, deterministic local actions, response audio, touchscreen GUI operation, physical Pi validation, integrity records, and a reproducibility package.

The honest engineering conclusion is not that E50 is near-perfect or production-grade. The evidence supports a narrower and stronger claim: E50 works as a frozen offline VCM prototype with strong tested action safety and fast local inference, while command-level recognition remains moderate across the full vocabulary. The final benchmark establishes 95.0% overall wake success, 68.42% raw command classification, 52.63% end-to-end action success, 100% accepted-action precision, 100% safe rejection among non-executed trials, 100% UNKNOWN/no-action safety, and 100% return-to-listening within the tested population.

The report deliberately preserves evidence boundaries. Broad environmental robustness, formal unseen-human-speaker Pi generalization, full acoustic end-to-end latency, full wake-to-response RTF, calibration, and broad noise/phrasing robustness were not established. E53 remains independent and does not alter E50. The project closes with its limitations visible rather than hidden.

20 Local laptop development and final E50 training evidence

The project evidence supports a continuous local laptop development path from the earliest baseline and trial stages through the final E50 command-model training record. The decision and experiment logs show that the first dataset inventory, preprocessing checks, sklearn smoke baselines, TensorFlow setup, E21 CNN baseline work, and laptop WAV / real-microphone trials were performed in the local Windows project environment during 2026-09-15–2026-09-16. Those early records describe local Python environments, locally available NumPy/SciPy/sklearn tooling, TensorFlow 2.20.0 installation and import verification, E21 training/evaluation artifacts, and laptop-recorded WAV trials used to diagnose real-microphone domain mismatch. The final E50 command model, `E50_BG_REVISED_VOCAB_ORIGINAL_PRESERVE_E41_INIT`, is separately supported by the Phase BG training records in the frozen project. The local filesystem evidence for the E50 training-result directory begins with `PHASE_BG_TRAINING_MANIFEST.csv` at 2026-09-28 01:29:43 and includes the trained E50 weights and normalization artifacts written at 2026-09-28 02:11:41; the final result and integrity artifacts extend through 2026-09-28 03:40:10. The Phase BG resource baseline states that the local BG run used CPU-only TensorFlow with Python 3.13.6, TensorFlow 2.20.0, and Keras 3.11.3; it also records that no TensorFlow GPU device was detected and that the TensorFlow CUDA build was false. The laptop hardware available for the project was a Think 15 B12UCX system named LOREENANNEDEGUZMAN, with a 12th Gen Intel Core i7-12650H processor, 16.0 GB installed RAM, NVIDIA GeForce RTX 2050 4 GB graphics plus Intel UHD Graphics, and 477 GB local storage. Although the laptop included a discrete GPU, the recovered TensorFlow evidence establishes the Phase BG/E50 run as local CPU-only execution, and no reviewed project evidence establishes A100-based final E50 training.

21 GitHub / Reproduction Package Guide

The GitHub package should be read as the detailed implementation and reproducibility bundle for this report:

1. Read `README.md` for overview, architecture, metrics, limitations, and reproduction path.
2. Inspect `01_REQUIREMENTS/` for traceability.
3. Inspect `02_FINAL_SYSTEM/` for deployable models, configs, runtime, router, GUI, and responses.
4. Inspect `03_ENGINEERING_HISTORY/` for decisions, experiment history, and exam notes.
5. Inspect `04_VALIDATION/` and `05_RESULTS/` for final benchmark, safety validation, timing, and limitations.
6. Inspect `06_REPRODUCTION/` for setup and run instructions.
7. Inspect `07_INTEGRITY/` for SHA256 manifest and artifact integrity evidence.
8. Inspect `09_INDEPENDENT_EXPERIMENT/` only for the static E53 narrative; E53 is not part of E50 deployment.